

Final Project Proposal: Stereo-Enhanced VLA: Liquid Pouring Robot with Depth Fusion

Team Members:

Zou Junfeng (225015057)

Cao Fengyu (225010398)

Ge Yuxuan (225040528)

February 6, 2025

Abstract

This proposal outlines the design and development of an intelligent robotic system capable of performing liquid pouring tasks in unstructured environments. Traditional robotic automation relies heavily on pre-programmed trajectories, which fail when facing natural language instructions such as "pour the Coke until the cup is half full." To address this limitation, we propose a progressive implementation strategy centered on Vision-Language-Action (VLA) models. As a fundamental task, we utilize a standard VLA model to achieve semantic object understanding and coarse-grained execution planning. Building upon this baseline, we introduce binocular depth fusion as an advanced enhancement to optimize manipulation performance and efficiency. Through this framework, we aim to validate the feasibility of VLA-driven robotics and evaluate the specific performance gains achievable through multi-modal depth integration. And the entire system will run on the Qualcomm RB8 development kit.

1 Project Overview

The field of Embodied AI has witnessed significant progress with the advent of Large Language Models (LLMs). Robots are no longer mere playback machines but are evolving into intelligent agents capable of understanding human intent. However, a significant gap remains between understanding a command and executing it physically, especially for delicate tasks involving fluids.

1.1 Motivation and Challenge

Pouring liquids is a canonical challenge in robotics because it involves complex fluid dynamics, transparent object perception, and continuous state estimation.

- **Fine-Grained Semantics:** Commands like "fill it halfway" require the robot to quantitatively understand volume, not just binary actions (open/close).
- **Spatial Limitation of VLAs:** While VLA models excel at semantic understanding, they inherently lack precise 3D spatial perception. Relying solely on 2D RGB input makes it difficult to infer exact depths and metric coordinates needed for safe grasping and pouring.
- **Edge Computing Constraints:** Running massive VLA models (like RT-2) typically requires cloud GPUs. Deploying these on a mobile robot platform (Qualcomm RB8) requires significant model quantization and optimization.

1.2 Project Objectives

The primary objective of this project is to develop a "Proof-of-Concept" system that demonstrates:

1. **Autonomous Recognition:** The robot identifies target objects (bottles, cups) from a cluttered table using VLA.
2. **Precise Manipulation:** The robot grasps the object without knocking it over by integrating binocular depth data.
3. **Volume-Aware Pouring:** The robot pours liquid and stops autonomously when the visual feedback indicates the target level is reached.

2 Technical Approach and Implementation Path

This project adopts a "Baseline-to-Advanced" progressive development strategy. We first establish a standard RGB-based VLA baseline to achieve a semantic control loop, and

then introduce an innovative depth fusion strategy to address the physical limitations of a single visual modality in spatial depth perception.

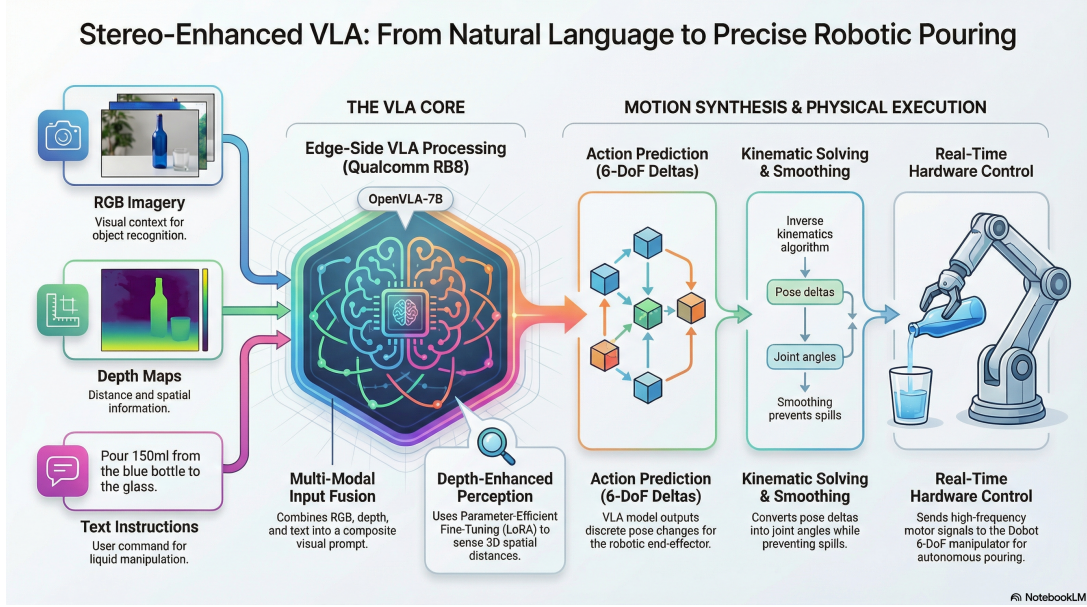


Figure 1: System Workflow: From User Instruction to Robot Execution.

2.1 Basic Strategy: Edge-Side VLA Control (Baseline)

This phase aims to build a system capable of understanding natural language and outputting action sequences, focusing on semantic understanding and cross-modal alignment.

- **Lightweight Model Deployment:** We will select a lightweight VLA model (e.g., OpenVLA-7B or a distilled RT-2) as the foundation. To adapt to the computational resources of the Qualcomm RB8 development kit, we will employ model compression and optimization techniques on the model backbone.
- **Standard I/O Stream:** The system receives a monocular RGB image stream and text instructions. The VLA model aligns visual features with language embeddings via Cross-Attention mechanisms, directly predicting the 6-DoF pose deltas of the robotic arm end-effector.
- **Limitation Analysis:** While the basic strategy excels at semantic generalization, the lack of absolute scale information in RGB images often leads to imprecise action prediction along the Z-axis (depth), making it difficult to perform fine alignment tasks directly.

2.2 Advanced Strategy: Depth Fusion with VLA model

To introduce precise geometric information without altering the Transformer input architecture of the VLA model, we propose a fusion scheme that "visualizes" depth infor-

mation.

- **Multi-Modal Fusion Input:** We fuse the raw RGB image with the depth map to synthesize a composite image input incorporating both “texture semantics” and “geometric depth.”
- **Fusion Fine-Tuning:** We perform Parameter-Efficient Fine-Tuning (LoRA) on the VLA using this composite image data. The model learns to perceive spatial distances from depth maps, thereby empowering the VLA model with the ability to sense precise spatial locations and significantly improving its performance and efficiency.

2.3 Execution Layer: Kinematic Solving and Motion Control

This layer bridges the gap between the high-level decision-making of the VLA and the low-level drive of the robotic arm, translating semantic action commands into smooth, executable motor control signals.

- **Pose Integration and Smoothing:** The VLA model outputs discrete end-effector pose deltas (ΔP) at a fixed frequency. The execution layer first accumulates these deltas to the current state to obtain the absolute target pose and applies interpolation algorithms to smooth adjacent waypoints. This step is crucial for eliminating discrete jitter from the model output, preventing liquid spillage caused by sudden acceleration changes during movement.
- **Inverse Kinematics (IK):** We construct an IK solver using the kinematic model of the Dobot arm. This module receives the smoothed Cartesian target pose and resolves it into six target joint angles ($q_1, \dots, q_6$). During the solving process, the system simultaneously validates joint limits and singularity configurations to ensure the planned configuration is physically feasible.
- **Hardware Setup:** For the robotic arm, we utilize the Dobot 6-DoF manipulator from the iHEAR Lab. With a payload capacity of 5kg, it is fully sufficient to implement this project. And the robot arm will be equipped with a gripper specialized for grasping and pouring tasks.

3 Key Technical Highlights

The key technical highlights of this proposal are outlined below:

3.1 Liquid Pouring via VLA

Traditional robotic pouring tasks often rely on hard-coded trajectory playback, failing to adapt to variable instructions. This project aims to innovatively apply Vision-Language-Action (VLA) models to the domain of object manipulation and liquid pouring tasks, targeting a direct mapping from natural language instructions to physical actions. We expect this to allow the robot not only to identify specific objects but also to comprehend abstract semantic concepts thereby endowing the robot with the cognitive intelligence to handle unstructured tasks.

3.2 Edge-Side Optimization and Deployment on RB8

We are committed to challenging the deployment of large models on resource-constrained edge devices. To run the 7B-parameter VLA model on the Qualcomm RB8 development kit, we plan to implement hardware-aware model compression strategies as well as inference acceleration targeting the Qualcomm AI Engine (NPU).

3.3 Depth-RGB Fusion Inference

To address the inherent deficiency of standard VLA models in spatial perception, we propose a Depth-RGB fusion inference strategy. By fusing depth information resolved from stereo vision with RGB images as visual prompts for the VLA model, the system is capable of translating explicit geometric distances into visual features comprehensible to the model. We anticipate that this approach will significantly improve the model’s spatial localization precision and manipulation accuracy.

4 Conclusion

This proposal presents a comprehensive plan to develop the “Stereo-Enhanced VLA: Liquid Pouring Robot with Depth Fusion.” To address the challenges of spatial perception and fine motor control in unstructured environments, we have established a progressive technical roadmap: evolving from a standard semantic control baseline based on VLA to an advanced inference architecture featuring Depth-RGB fusion.

This project is dedicated not only to validating the feasibility of VLA models in complex, long-horizon tasks such as “object manipulation and liquid pouring,” but also to challenging the extreme optimization and deployment of large models on the Qualcomm RB8 edge computing platform. By organically combining the breadth of semantic understanding with the precision of geometric perception, we aim to create an intelligent system capable of both comprehending abstract instructions like “pour half a cup” and executing precise physical actions.